

CASE REPORT**PATHOLOGY/BIOLOGY**

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Sudden Death Due To Acute Cocaine Toxicity —Excited Delirium in a Body Packer

ABSTRACT: Excited delirium denotes a life-threatening medical condition characterized by the acute onset of agitated and violent behavior that often results in a sudden and unexplained death. Cocaine-induced excited delirium refers to fatal cocaine intoxication with the following symptoms occurring sequentially: hyperthermia, delirium with agitation, respiratory arrest, and death. We present a case of cocaine-induced excited delirium in a cocaine “body packer” or a “mule”, specifically an individual who attempts to smuggle cocaine within the body. Investigators at the scene initially suspected homicide due to the victim’s sharp and blunt force injuries. Three rubber packets containing cocaine were removed from the victim’s rectum. Blood toxicological analysis revealed an alcohol concentration of 0.016 g/100 and cocaine >1 mg/L. The forensic pathologist should consider cocaine-induced excited delirium when an individual exhibits aggressive behavior, unexpected strength, and resistance to pain who dies suddenly. Further analysis should be performed during the scene investigation and autopsy for evidence of body packing.

KEYWORDS: forensic science, forensic pathology, cocaine, excited delirium, sudden death, body packer

Initially described in 1849 by the psychiatrist Dr. Luther Bell, “Bell’s Mania” refers to individuals with a host of symptoms, including sudden onset of overactivity, marked sleeplessness, disconnected delusions, and hallucinations (1,2). Also known as “lethal catatonia”, “acute exhaustive mania”, and “agitated delirium”, this diagnosis was primarily attributed to psychiatric patients. In 1985, Wetli and Fishbain coined the term “excited delirium” to describe cases of fatal cocaine intoxication with the following symptoms occurring sequentially: hyperthermia, delirium with agitation, respiratory arrest, and death (3,4). There has been a significant increase in excited delirium fatalities reported since the late 1980s, attributed to the burgeoning popularity of crack cocaine (4,5).

The lifetime prevalence of recreational cocaine use is currently estimated to be >14% in the United States (6). Approximately 1.2% of Europeans used cocaine in 2007, ranging from 0.7% in Romania and Lithuania to 12.7% in the United Kingdom (6). Lucena and colleagues reported that a total of 76% cocaine-related sudden death victims had detectable serum ethanol concentrations (6). Cocaine and ethanol potentiate each other, leading to cardiotoxicity exemplified by myocardial depression, decreased coronary arterial blood flow, dysrhythmias, and sudden death (7).

The National Association of Medical Examiners (NAME) published a position paper in 2004 on the certification of cocaine-related deaths (8), which has since sunsetted and not been reinstituted. However, the considerations in that position paper are still generally accepted among medicolegal death investigators. The committee stated that the investigation of any drug-related death requires a complete analysis of the circumstances of death, the death scene, and past medical history. Furthermore, it was also mandatory to obtain the results of the forensic toxicology examination coupled with a complete forensic autopsy prior to determining the cause and manner of death.

Excited delirium denotes a life-threatening medical condition characterized by the acute onset of agitated and violent behavior that often results in a sudden and unexplained death (9–12). The classic case of excited delirium consists of a man who is excessively agitated or belligerent, resistant to pain, and hyperthermic (temperatures of 104–107°F are common). Other signal components include incoherent speech, hallucinations, history of substance abuse or psychiatric disorders, and tachycardia (4,13–17). Many are naked or underclothed and are attracted to water, lights, or glass. Men with excited delirium account for 10% of cocaine deaths in Miami (4). Furthermore, approximately 39% of cases of fatal excited delirium in Miami transpired in police custody (4).

We report a case of cocaine-induced excited delirium in a cocaine “body packer” akin to a “mule”; specifically an individual who attempts to smuggle cocaine within a body orifice. We discuss sudden death due to cocaine-induced excited delirium and the common findings discerned at autopsy as well as the characteristics of a “body packer.”

Case History

A 38-year-old white male was found dead lying nude on the floor of his ransacked hotel bathroom. Hot water was running

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from the bathtub tap, and several inches of water had overflowed the bathtub. Blood was noted at the scene mixed with the water on the bathroom floor. The decedent had entered the hotel room alone the evening before his discovery. Witnesses heard a commotion in the room during the night; however, the victim was the only individual observed near the scene. Water was seen in the hotel hallway the next morning coming from his room.

Upon police entry, the room was in a state of disarray with damaged light fixtures, overturned bedding, and a smashed toilet. A partially eaten hamburger was observed in the room. Toilet paper was stuffed under the bathroom door from the inside of the bathroom. A hammer was found in the bathtub, leading investigators at the scene to initially suspect homicide as the manner of death. Drug paraphernalia was noted in the hotel room including a syringe with a bent needle. The decedent had a history of cocaine abuse as well as undressing and acting violently.

Autopsy Findings

The decedent was a well-developed, well-nourished male measuring 5 feet 4 inches long and weighing 167 pounds. The external examination was significant for multiple irregular to parallel oriented scars of the anterior and posterior surfaces of the upper and lower extremities. No needle tracks were identified; however, punctate injuries were on the left arm. Irregular scarring and a brown contusion were in the left antecubital fossa.

Multiple incisions covered the arms, neck, anterior chest, anterior surface of the legs, feet, buttocks, upper back, shoulders, face, and vertex of the head. None of the injuries penetrated vital organs within the trunk or skull. There was no intracranial hemorrhage and no injuries of thoracoabdominal viscera. The hands had multiple contusions and incisions over the palms and dorsal aspects. Numerous wounds of the feet, legs, arms, back, and head contained firm, gray/white jagged fragments resembling porcelain (Fig. 1).

Within the rectum, there were three tan, latex-covered packets, which were tied at one end and round at the other end (Fig. 2). They measured from $\frac{3}{4}$ inch to 1 $\frac{1}{8}$ inch in length and from $\frac{5}{8}$ inch to $\frac{3}{4}$ inch in diameter (Fig. 3). Two packets were soft and pliable with a firm material within them. The smallest packet was rough and irregular and had a triangular-shaped tear on the bulbous end exposing a white plastic material.

There was slight cardiac hypertrophy (heart weight: 410 g) with mild left ventricular thickening as well as pulmonary edema (right lung: 630 g; left lung: 530 g) and cerebral edema.



FIG. 1—Multiple incisions of the arms. The arrow depicts a firm, gray/white jagged fragment resembling porcelain.

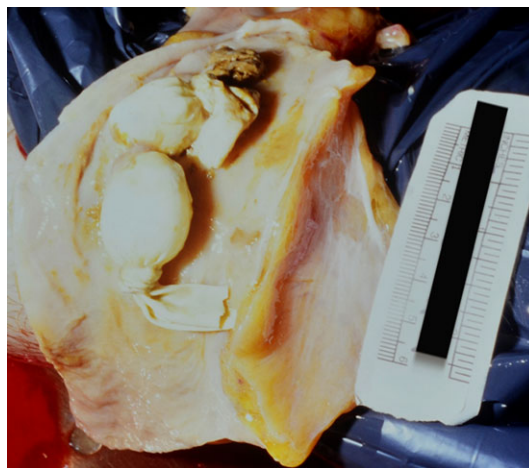


FIG. 2—Three latex-covered packets were found in the victim's rectum. The packets were tied at one end and round at the other end.



FIG. 3—The three packets found in the victim's rectum.

Toxicological Analysis

Blood toxicological analysis revealed an ethanol level of 0.016 g/100 and cocaine at >1 mg/L. The contents of the three rubber packets removed from the rectum underwent toxicological analysis. White powder had been placed in clear plastic resembling the finger of a latex rubber glove and then tied upon itself. One of the bags contained a small hole in the outer rubber cover but showed no obvious moisture inside. The weights of the three bags were 1.582, 1.625, and 1.595 g. Each of the packets contained cocaine, as did the syringe found at the scene.

The cause of death was acute cocaine toxicity—excited delirium. The manner was accident. The decedent had not traveled internationally prior to his death and was simply storing his personal “stash” of cocaine within his rectum.

Discussion

Several studies in the literature have addressed the phenomenon of cocaine excited delirium (3,5,16,18–23); however, a paucity of reports have involved a cocaine “body packer” (23–31). In Wetli and Fishbain's landmark study of sudden death and cocaine-induced psychosis, numerous common features were noted among the seven cases (3). Six of the seven victims demonstrated unexpected strength, requiring several people to restrain them. Physical restraints were utilized in six cases. Hyperthermia was observed in four cases, and mydriasis was seen twice.

Cocaine excited delirium deaths often occur in police custody (5,18–21,32,33). In Ross' study of 61 cases of excited delirium fatalities in police custody, all of the victims fought with and were restrained by police, the victim was more likely to die at the scene or during police transport, and police were commonly responding to a disturbance call (21). Survival time was <1 h in many cases. The most common cause of death was acute cocaine toxicity, while physical restraint in police custody was contributory.

Grant and colleagues analyzed 202 custodial deaths between 1939 and 2004 in Maryland (32). Of note in their study, sudden unexplained deaths in custody included violent and/or erratic behavior, the use of numerous restraints, and drug intoxication (32). This triad of factors was not recognized until the 1980s which corresponded to an era of increased cocaine abuse. From the 1980s until the conclusion of their study, the authors noted that the victims were younger, the manner of death was usually undetermined, and the fatalities more likely involved drug intoxications and restraints (32).

Pollanen and colleagues reviewed 21 cases of unexpected death in individuals with excited delirium (20). A total of 18 victims (86%) were in police custody at the time of death. All of the 21 cases were associated with restraint for agitation or hyperactivity, with 18 victims (86%) in the prone position and 3 (14%) having pressure applied to the neck (20). Excited delirium was attributed to a psychiatric disorder in 12 cases (57%) and to cocaine-induced psychosis in 8 cases (38%). O'Halloran and Lewman reported 11 cases of restraint asphyxiation in police custody (18). All of the victims were in an excited delirious condition when restrained in a prone position: 9 were hogtied (binding a person's wrists and ankles behind his/her back while he/she lies prone); 1 was tied on a gurney, and 1 was manually held prone by several police officers. Three of the victims were psychotic, while the others were acutely delirious due to drugs (6 from cocaine, 1 from methamphetamine, and 1 from LSD.) One victim undressed, while at least 3 others broke glass.

A "body packer" or "mule" is an individual who smuggles illicit drugs by intracorporeal methods for the purpose of evading law enforcement (25,29,30). A host of drugs have been transported through these means, including cocaine (23–30,34,35), methamphetamine/amphetamine (36–38), opiates (27,30,39), and phenacetin/cocaine (40). These smugglers wrap the illicit drugs in plastic bags, balloons, or condoms, and subsequently swallow or insert the items into a body orifice (25). The most common cause of death among body packers is acute drug intoxication due to disintegration of the packages within the gastrointestinal tract (29). Death attributed to acute cocaine toxicity may ensue without rupture of the packets if the packets consist of semipermeable membranes (23).

In the majority of cocaine "body packer" cases, the individual retrieves the packets with use of laxatives and/or enemas after clearing U.S. Customs (25). Rupturing of endoluminal packets in the gut typically prompts an emergency room visit (24,28,30,31,34) or causes sudden death (25–27,29,31). In their study of 1,250 confirmed "body packers" apprehended at U.S. Customs at JFK airport in New York, Mandava et al. reported that 70% of "body packers" were men, with a male to female "ratio of 2.8 to 1" [sic] (30). The mean age was 33 years, and 52% were from Colombia. Heroin was the most common illicit drug transported (73% of cases). A total of 25 (45%) patients underwent surgery, while the remainder was managed conservatively.

Several studies in the literature have addressed the phenomenon of cocaine "body packers" who succumbed to sudden death (Table 1). All of the victims had smuggled cocaine in a

body orifice while traveling internationally. Most of the decedents had traveled from South America to the United States, with some dying during the flight and others at home, in a hotel room, or at the emergency room. A variety of symptoms was noted, including extreme polydipsia, agitation, seizures, acute toxic psychosis, and coma (23,25). The majority ingested a balloon, condom, or plastic bag filled with cocaine, while others inserted the packet of cocaine in their rectums (23,31). One woman was discovered with two double-wrapped plastic bags filled with 170 g of cocaine in her vagina (23).

The scene investigation may offer clues as to the cause or manner of death in cases of body packers. There may be evidence of laxatives, enemas, or tickets suggesting a recent international flight (23). An ice pack or wet towel may be present suggesting premortem hyperthermia. Of note in the present case, hot instead of cold water was running from the bathtub tap, which eventually overflowed the tub.

Although the manner of death in most body packer cases is accidental, a suicide attempt and completion have been reported (23,25,35). Stichenwirth et al. reported the case of a 30-year-old Colombian man who traveled from Bogota to Vienna and who was arrested for suspicion of drug smuggling (35). The body packer admitted to ingesting approximately 80 cocaine packages. The victim expelled 59 packages aided by a laxative, after which he was transferred to prison and placed under constant surveillance. Prison nurses only performed hourly checks on the victim. He was found dead in his cell 23 hours after apprehension. Eight open and empty packages were located near the decedent. Toxicological and autopsy findings implied that eight expelled packages had been opened by hand and swallowed with suicidal intention.

Fishbain and Wetli described a 31-year-old man who presented to an emergency department with symptoms of agitation, hyperactivity, confusion, disorientation, and incoherence (23,25). He was restrained by six men. Within 3 ½ hours after arrival,

TABLE 1—Cocaine "body packers" in the literature.

Studies	Year of Study	Deaths	Travels
Shields et al. Present Study KY N = 1	2014	1	None
Fishbain et al. (25) FL N = 1	1981	1	Columbia → U.S.
Wetli and Mittleman (23) FL N = 11	1981	10	Columbia → U.S. (N = 10) Bolivia → U.S. (N = 1)
Furnari et al. (26) Rome, Italy N = 1	2002	1	Columbia → Rome
Suarez et al. (31) FL N = 3	1977	2	Columbia → U.S. (N = 2) Unspecified international travel (N = 1)
Koehler et al. (29) PA N = 1	2005	1	Jamaica → U.S.
Stichenwirth et al. (35) Austria N = 1	2000	1	Columbia → Amsterdam → Vienna

the victim was in respiratory arrest and died. Autopsy findings demonstrated an empty aluminum foil packet in the pylorus, a ruptured rubber finger cot in the cecum, and marked hemorrhagic pulmonary edema. Police investigation determined that the victim had returned from Colombia 10 days earlier and was a cocaine body packer. He had passed almost all of the finger cots of cocaine when he developed cocaine toxicity, at which time he swallowed aluminum foil as a suicide attempt to seek medical assistance and to avoid a police investigation.

The investigators in the present case initially suspected homicide as the manner of death due to the sounds of a commotion in the hotel room before the victim's death and the state of disarray in the room, including damaged light fixtures, overturned bedding, and a smashed toilet. In addition, a hammer was discovered in the bathtub. The victim also had evidence of incisions on numerous areas of his body, none of which punctured vital organs. The manner of death was determined to be accident following a thorough investigation and complete autopsy, which confirmed cocaine excited delirium.

Autopsy findings of victims of cocaine-induced excited delirium are often minimal and nonspecific (4). The decedent may have evidence of self-inflicted injuries due to cuts from broken glass and may have abrasions, contusions, or lacerations attributed to restraints, physical fights with police (17), or scratches from formication (i.e., tactile hallucinations of insects crawling underneath the skin) (41). Postmortem blood cocaine concentrations are usually <1 mg/L, while there are often high blood levels of benzoylecgonine, the principal cocaine metabolite (3,17). Postmortem toxicological analysis of fatal cocaine-associated excited delirium cases demonstrated cocaine concentrations similar to those seen in recreational drug users, and less than those found in acute cocaine intoxication deaths (42). In the absence of packets recovered in the gastrointestinal tract or body orifice, the autopsy findings of a cocaine "body packer" are consistent with those of a drug overdose (23). Nonspecific autopsy findings include pulmonary congestion and edema, cerebral edema, and gastric hyperemia.

The decedent in the present case displayed features of both cocaine excited delirium and a body packer. The cocaine excited delirium was exemplified by an individual with a history of cocaine abuse, the destruction in his hotel room, the running water in the tub, the self-inflicted nonfatal incisions on his body, and the presence of fragments of white porcelain-like material on his legs, back, and head. The discovery of three rubber packets in his rectum each containing >1.0 g of cocaine convincingly undergirded the conclusion that he was a body packer. Cocaine was present in the blood at a concentration of >1 mg/L. A syringe with a bent needle containing cocaine was located near the body. The entire literature of body packers depicts an individual who smuggled an illicit drug internationally to evade law enforcement. To our knowledge, we present the first published fatality of a body packer, who protected his personal "stash" of cocaine within the gut in an attempt to avoid police capture and then succumbed to cocaine excited delirium.

Conclusion

The National Institute of Justice of the United States of America developed a Technology Working Group in April 2011 consisting of experts in the area of excited delirium syndrome (42). Due to the overlapping behaviors and clinical signs between excited delirium and other medical conditions, excited delirium has been labeled as a syndrome. It has been reported

that disturbances of dopamine neurotransmission may account for the profound hyperthermia noted in excited delirium patients (42). With the increase in cocaine use in the 1980s, cocaine-induced excited delirium is often observed in the setting of a restrained individual in police custody who succumbs to a sudden and unexpected death (25,42). Further investigation should be conducted in these cases to determine the source of the cocaine, which may be secretly hidden within the decedent's body. We report the sudden death of a victim of cocaine-induced excited delirium, who—unlike the classic "mule"—had surreptitiously placed his personal "stash" of three packets of cocaine within his rectum to evade police detection.

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